Energy-Flow Feasibility and Learned Redundancy Management for Persistent K-Coverage UAV Networks

Professor progress meeting — complete context

Himanshu Nagwanshi, Surya Prakash S, Harsh Varshney, Gaddam Viswavijith

Agenda

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The blank

UAV coverage papers report a number and stop. A coverage ratio without an achievability reference cannot distinguish a strong controller from an infeasible instance; a binary "all constraints hold now" metric is identically zero for every policy on an infeasible instance. Worse, in persistent operation the instance can be infeasible in *energy* while still feasible in *space*. This project fills the blank: a feasibility certificate that says, before any controller is measured, whether the instance can be covered at all, whether energy sustains it, and over what horizon.

The system under test

World

- $N = 8$ active UAVs + 1 reserve; $M = 20$ sensors in 80×80 m
- $K = 2$ redundancy target; 2 charging pads
- Actions: hover / 4 moves / charge; step 0.2 s at 5 m/s
- Start: connected cluster in one corner; covers 3/20 sensors
- Mission: 2000 steps = 400 s

What makes it hard

- Coverage wants UAVs spread; connectivity wants them close; energy wants them docked
- A UAV moves toward a deficit and can strand the far side
- 8 UAVs draining at the same rate queue on 2 pads
- The deployed policy sees only local observations, no global state

The configuration (Table VII, report)

N, M, K	8+1, 20, 2
Field Ω	80×80 m
R_{cov}, R_{comm}	35 m, 40 m
Altitude z_0	50 m
$B^{\max}$	30,000 J
Hover / Move / Avionics	50 W / 75 W / 5 W
Charge rate	0.0002 per step
Episode	400 s (2000 steps)

The centred-ring alternative covers 17/20 sensors at $t = 0$ ($CK = 0.850$ floor), so no controller is distinguishable. The corner cluster keeps the comparison separable: the controller must deploy across the field while holding $K = 2$. Every row below every table in the report means the same frozen policy on the same 20 held-out scenes (seeds 1000+), 2000 steps.

Old report → current report

	Old (10 pp)	Current (25 pp)
Claim	deferred	measured with intervals
Result	none verified	0.9512 vs 0.923; classical 0.929
Wins / losses	unknown	low-comm and failures win; energy and start lose
Persistence	assumed	not feasible; 400 s horizon only
Metric	SC (degenerate)	graded metrics + refresh time
PHY	asserted	INET-verified 0.012 dB

Ten pages of intent became 25 pages of measurement; nothing was asserted without an artifact the claims ledger points to.

What the report now reports

The result is not "learning wins". It is a *split*: the frozen policy beats the strongest rule by $+0.028$ on nominal (0.9512 vs 0.923, interval excludes zero), by $+0.125$ on low connectivity, and by $+0.034 / +0.035$ on single and multi failure. It loses by -0.064 where batteries are scarce and -0.038 where the start is spread. A single "does learning beat the rule" question would hide exactly what the certificate separates. The energy-flow ceiling then explains the time horizon of every number above.

The certificate (spatial, Section III-G)

Area surplus is not a covering argument: $N\pi R^2/KA = 1.503$ passes at $N = 5$, yet the four corner points are 80 m apart, more than $2R_{\mathrm{cov}} = 70$ m, so no disc serves two corners. Hence $N \geq KP(A, R_{\mathrm{cov}})$, and for the region $N \geq 4K = 8$ with equality achievable. For a drawn sensor set the packing number has mean 2.13, so $N = 5$ is excluded on 13% of draws. The teleporting oracle reaches 1.000 at $N = 8$, so the instance is coverable; the service-continuity metric is identically zero when $\mathcal{F} = \varnothing$ and is replaced by time-averaged deficit and maximum refresh time.

The energy certificate (Section III-H)

Each pad returns 24.9 W net (30.0 W gross minus 5.1 W avionics). One airborne UAV draws 55.1 W. Steady state sustains 0.90 airborne UAVs against 8 required, so the persistent ceiling is 0.113. Finite horizon: $T = 400$ s gives ceiling 1.000 by spending the initial packs; $T = 2000$ s gives 0.367 at full charge and 0.326 at mean initial charge.

Only $N = 12$ with 4 pads, 60 kJ packs and a 0.0004 charge rate reaches steady state. Energy-flow is therefore a hard constraint the spatial certificate cannot see, and it bounds every long-horizon reading of the system.

How the learner is built (Section IV)

Parameter-shared MAPPO, centralized critic at training, decentralized actors at deployment. The actor sees 68 dimensions: 30 base proprioceptive terms, a 4×4 deficit map (16), one-hot identity (8), pad bearing (3), ego-centric bearings to the $k = 3$ nearest under-covered sensors with neighbour repulsion (11). All six actions are masked by a safety filter with a 3-step connectivity lookahead and articulation-point exclusion. The reserve UAV re-seats a failed slot; a pad scheduler arbitrates charging. Training samples actions; deployment samples too, not the argmax.

The head-to-head (nominal, held-out)

Controller	CK	Notes
Weakest rule (fixed formation)	verifies the gap to ceiling = 0.628	
Strongest rule (Reactive-local)	0.923	same 20 scenes
Classical OR scheduler	0.929	$sd = 0.062$; Hungarian + oracle formation
MAPPO (68-dim, ego+shaping)	0.9512	[0.938, 0.964]; 5 seeds
Oracle ceiling	1.000	teleporting placement

Paired difference learner minus rule $+0.028$ with interval $[+0.016, +0.041]$; the classical scheduler was built to close the "weak baseline" objection and still trails by 2.2 points.

The pattern holds under stress

Regime	Rule	MAPPO	Classical
Nominal	0.921	0.947	0.929
Small battery	0.829	0.765	0.791
Low comm	0.804	**0.928**	0.808
Spread start	0.997	0.959	0.980
Failure	0.912	0.946	0.919
Multi-failure	0.910	0.945	0.919

Frozen policy, zero-shot. Wins sit exactly where adaptation and coordination matter (connectivity, faults). Losses sit where explicit energy conservation or deployment geometry dominates. The certificate is what lets us say which is which.

The learning experiment: planning and distillation

Planning is split into two steps. Generation: for every step the actor samples $K = 4$ candidate joint actions; each is rolled out $H = 2$ steps in a *cloned global* simulator and scored by cumulative coverage; the best executes. Distillation: those 200,000 (local observation, chosen action) pairs train the actor by behaviour cloning (Adam 1e-4, cross-entropy, 20 epochs). At deployment the distilled actor sees only the local 68-dim observation, one forward pass, no simulator, no critic. One signal: planning as an oracle gives $+0.0211$ resolved; distilled gives $+0.0106$ directional, recovering about half the oracle gain without the privileged state.

Verification: audit and INET

Claim audit

- 38 items: 34 pass, 4 informational, 0 fail
- Recomputes packing bound, oracle construction, determinism (170 identical values), and PHY path loss
- PHY: Python log-distance equals INET BreakpointPathLoss to 0.012 dB over 24 points

INET execution

- OMNeT++/INET 4.6: 8 UAVs, 20 sensors, BonnMotion trace
- Headless run: $t = 40$ s, event #71885, End; scalars in `General-#0.sca`
- PDR at mission scale is geometry/battery-bound, not radio-bound

What the project contributes and what it does not

It contributes a spatial certificate that proves N=5 infeasible, an energy-flow bound that proves the benchmark has a 400 s horizon, a teleporting oracle that separates policy failure from instance infeasibility, and a frozen-policy protocol in which learning demonstrably wins on connectivity and faults and honestly loses on energy and start geometry. The classical baseline and the INET fidelity check make those comparisons harder to dismiss.

It does not contribute a new RL algorithm, a hardware-validated deployment, or an indefinite persistent mission. Planning is an oracle upper bound; the deployable gain is smaller and directional. The novelty is the measurement discipline, not the method inside it.

What remains open (one question)

Three honest limitations remain. The config is not persistently feasible, so "persistent" names the 400 s horizon, not eternity. The distilled gain is directional, not resolved. And no hardware validates the simulator. The open question that follows is physical, not statistical: if the battery wave synchronizes (all 8 UAVs hitting the return threshold together on 2 pads), can a policy that recharges *early* and staggers the fleet break the small-battery wall that both the rule and the classical scheduler hit at 0.83? That experiment is the next step, not a formatting change.